

SAC vs SAC+JEPA with a cosine-normalized one-step objective

A 10,000-transition RTX 3090 comparison

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Summary. Normalizing the predictive loss to a scale-free cosine distance fixes the loss-scale defect of the earlier MSE branch and yields an auxiliary objective that is demonstrably learning: at 10k transitions the predictor reaches 0.0038 against a shuffled-action loss of 0.0099 and a no-model identity baseline of 0.1253. Performance, however, remains below vanilla SAC — 1.188 versus 1.655 — in this single-seed, 10k-transition comparison.

1 Shared setup

All three arms use identical terrain (standardized centered-Gaussian, periodic tiling, random start positions), the $P_{\max} = 6$ longitudinal power cap, 16 parallel environments, the same replay configuration (20,000 capacity, batch 128, 1,000 learning-start transitions, one update per transition), the learned entropy temperature, the same evaluation interval of 2,048 transitions, and held-out terrain seeds 100001–100004. Only the actor objective differs. Each arm is a single training seed; the reported $\pm$ are standard deviations across the four held-out terrain seeds, not across runs.

2 Full objective

The actor minimizes

$$\mathcal{L}_{\text{total}} = \underbrace{\mathbb{E}_{o \sim \mathcal{D}} [\alpha \log \pi_{\theta}(a|o) - \min(Q_1, Q_2)(o, a)]}_{\text{SAC actor loss}} + \lambda_{\text{JEPA}} \mathcal{L}_{\text{JEPA}} + c_{\text{var}} \mathcal{L}_{\text{var}}, \quad a \sim \pi_{\theta}(\cdot|o), \quad (1)$$

with the scale-free predictive term

$$\mathcal{L}_{\text{JEPA}} = 1 - \cos\left(q_{\psi}\left([f_{\theta}(o_t), a_t]\right), \text{sg}\left[\bar{f}(o_{t+1})\right]\right), \quad (2)$$

and the VICReg-style variance floor

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{j=1}^d \text{relu}\left(c_{\text{var}}^{\text{tgt}} - \sqrt{\text{Var}_b[f_{\theta}(o_t)]_j} + \epsilon\right), \quad \epsilon = 10^{-4}. \quad (3)$$

Here f_{θ} is the online actor encoder, q_{ψ} a $130 \rightarrow 128$ predictor consuming the current embedding and the executed action, $\bar{f}$ a stop-gradient EMA copy of f_{θ} , and sg the stop-gradient. SAC’s target *critics* use $\tau = 0.005$; this EMA target is separate and updates once per 64 collected transitions with $\tau = 0.01$. Replay supplies the executed tanh-squashed action and the true pre-reset next observation. Because $\cos$ discards magnitude, $\mathcal{L}_{\text{var}}$ is what prevents collapse; the two terms are complementary, not alternatives. Substituting MSE for the cosine gives the “corrected MSE” arm.

Setting	SAC	SAC+JEPA (MSE+var)	SAC+JEPA (cosine)
λ_{JEPA}	0	0.025	0.025
predictive loss	—	MSE on embeddings	$1 - \cos$
c_{var} / target	—	0.01 / 0.1	0.01 / 0.1
EMA cadence	—	per 64 transitions	per 64 transitions

3 Results

Transitions	SAC	SAC+JEPA (MSE+var)	SAC+JEPA (cosine)
2,048	0.064 ± 0.031	0.065 ± 0.030	0.064 ± 0.029
4,096	0.190 ± 0.027	0.174 ± 0.028	0.275 ± 0.035
6,144	0.292 ± 0.042	0.286 ± 0.036	0.269 ± 0.027
8,192	0.673 ± 0.041	0.701 ± 0.033	0.651 ± 0.030
10,000	1.655 ± 0.056	1.403 ± 0.020	1.188 ± 0.042

The endpoints are ordered monotonically in the influence of the auxiliary term: no JEPA (1.655) > MSE JEPA (1.403, -15.2%) > cosine JEPA (1.188, -28.2%). All three arms share an effectively identical entropy trajectory (α : 0.1457, 0.0788, 0.0426, 0.0231, 0.0135) and near-identical critic loss, so the divergence is late and confined to the policy representation.

4 The objective is learning, not collapsing

The cosine arm logs three quantities per checkpoint that separate a working predictor from a degenerate one: the actual loss, the loss with actions permuted across the batch, and the no-model identity baseline $1 - \cos(z_t, z_{t+1})$.

Transitions	$\mathcal{L}_{\text{JEPA}}$	shuffled action	identity baseline	$\lambda \mathcal{L}_{\text{JEPA}}$
2,048	0.00801	0.01863	0.26834	2.0×10^{-4}
4,096	0.00355	0.01167	0.35455	8.9×10^{-5}
6,144	0.00292	0.01080	0.19969	7.3×10^{-5}
8,192	0.00264	0.00944	0.15826	6.6×10^{-5}
10,000	0.00376	0.00986	0.12528	9.4×10^{-5}

At 10k the predictor is $33\times$ better than the identity baseline and $2.6\times$ better than the action-shuffled variant, so it is neither collapsed nor ignoring the action. The action-conditioning is nonetheless secondary: the shuffled predictor already beats identity by $12.7\times$, so most of the predictive power comes from z_t alone.

Two caveats bound the interpretation. First, the weighted auxiliary contribution $\lambda \mathcal{L}_{\text{JEPA}} \approx 9.4 \times 10^{-5}$ is 0.003% of the actor loss (-3.304); a loss this small relative to the policy term can still carry a disproportionate *gradient* because the cosine gradient scales with $1/\|z\|$, so loss magnitude alone does not establish that the term is inert. Second, the variance floor is active only at the first checkpoint ($\mathcal{L}_{\text{var}} = 0.011$, then 0 thereafter, with embedding standard deviation 0.09–0.12 above the 0.1 target), so the comparison is effectively SAC-only versus MSE-JEPA versus cosine-JEPA at $\lambda = 0.025$.

5 Performance plots

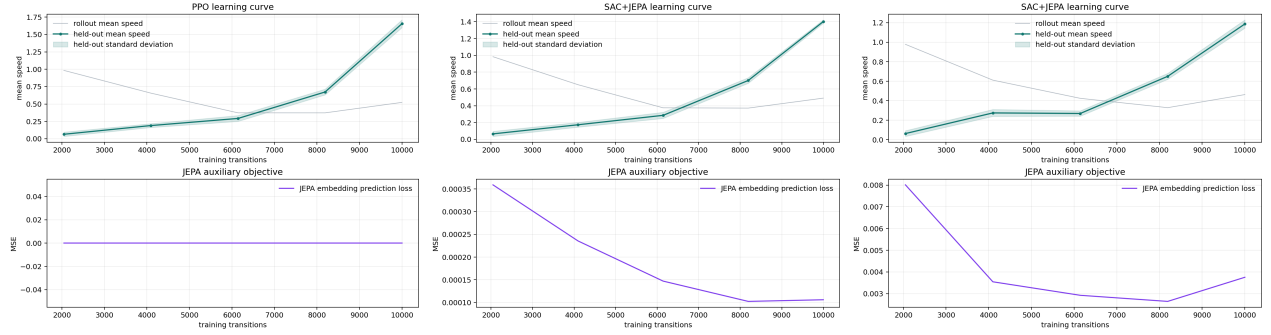


Figure 1: Learning curves. Left: SAC. Centre: SAC+JEPA MSE+variance. Right: SAC+JEPA cosine. Lower panels are the auxiliary objective.

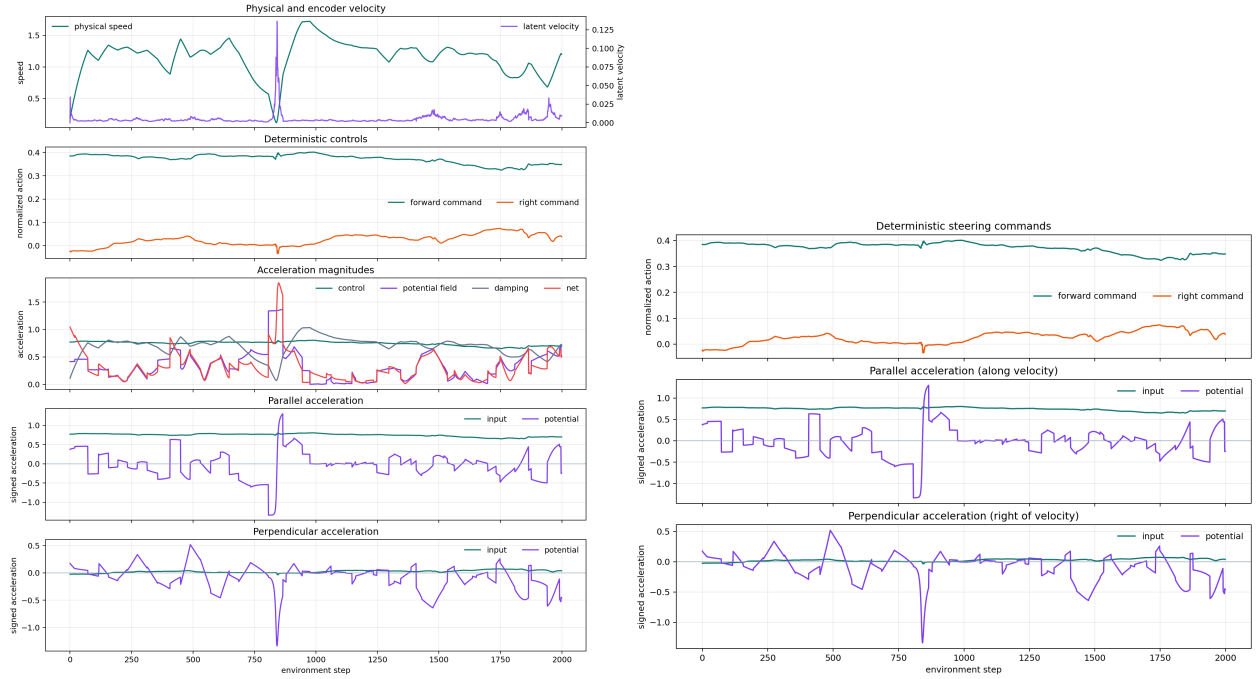


Figure 2: Cosine arm, held-out seed 100001: physical and latent velocity with deterministic controls, and acceleration magnitudes with signed parallel/perpendicular components.

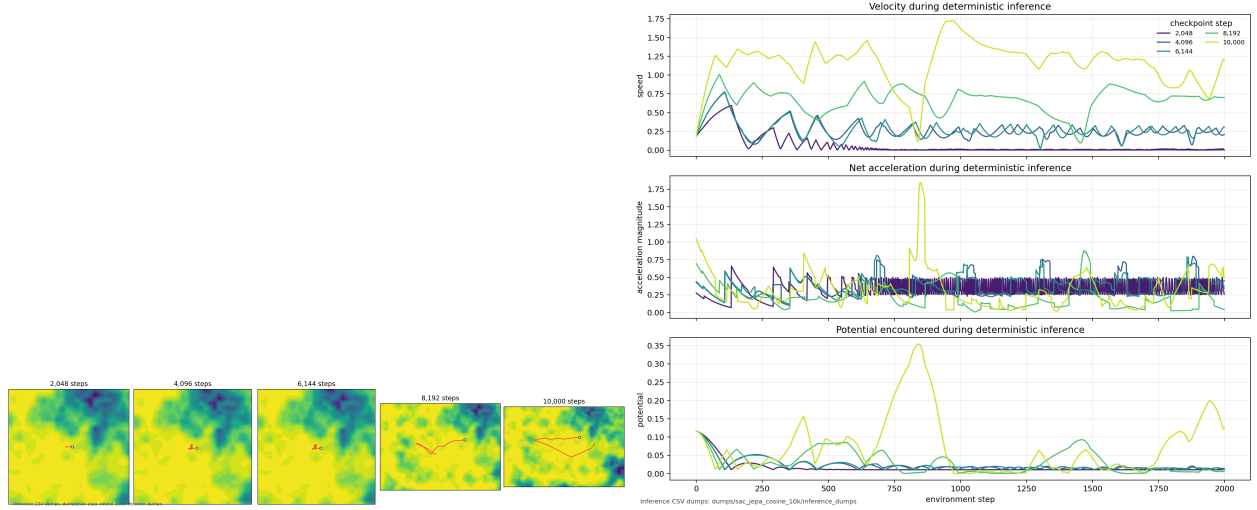


Figure 3: Cosine arm: deterministic inference trajectories over the terrain and the corresponding state/action histories.

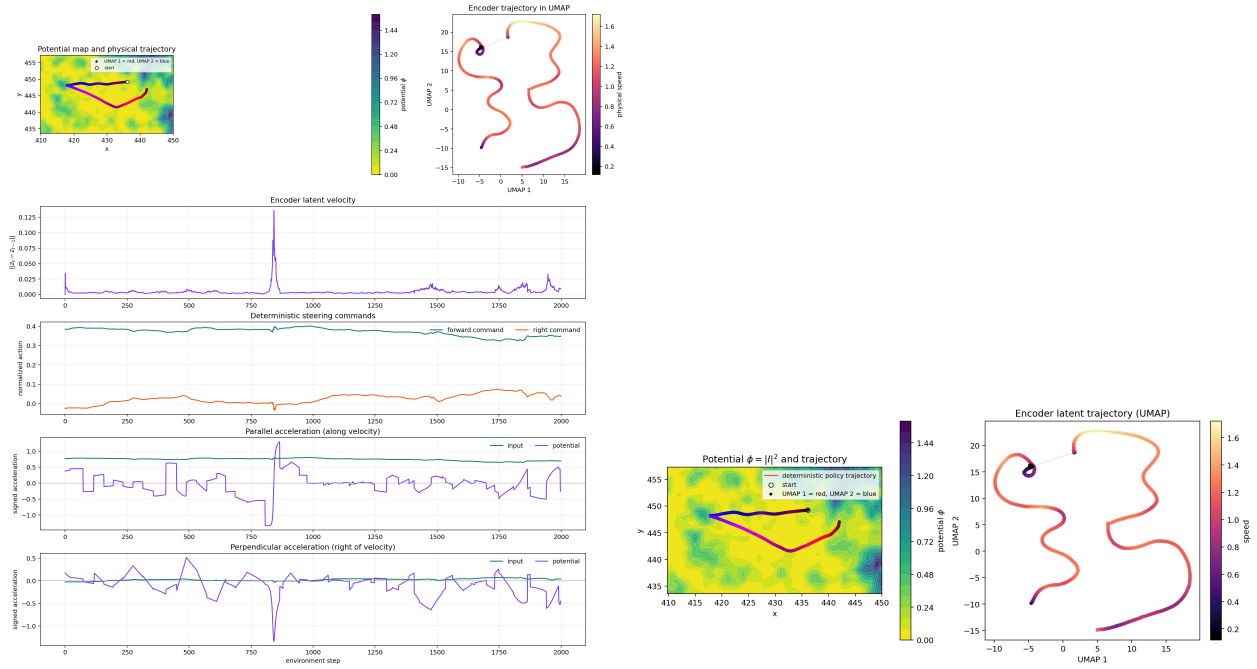


Figure 4: Cosine arm: full episode diagnostic and endpoint actor-encoder UMAP coloured by physical speed.



Figure 5: Cosine arm: pairwise joint UMAP grid across checkpoints 2,048 through 10,000 on one common held-out trajectory.

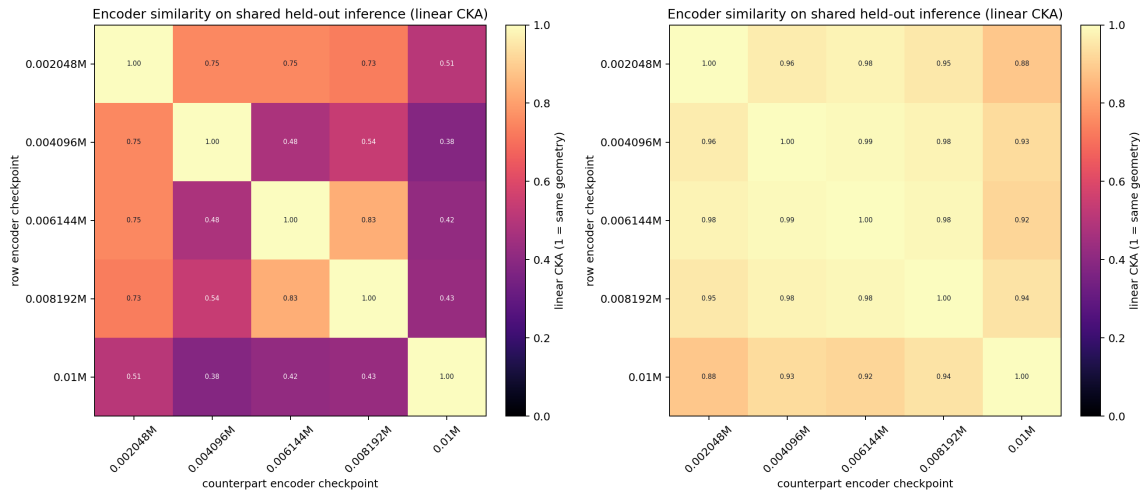


Figure 6: Linear CKA between checkpoint encoders on a held-out trajectory. Left: SAC. Right: SAC+JEPA cosine.

6 Conclusion

The loss-scale defect is fixed. Cosine normalization makes λ a scale-free weight, the adversarial diagnostics show a predictor that genuinely uses both z_t and a_t , and the objective is no longer the trivial near-identity map diagnosed in the earlier MSE branch.

The auxiliary objective nevertheless still trails vanilla SAC at this budget: 1.188 against 1.655 (-28.2%), with the MSE variant intermediate at 1.403. Three limits prevent a stronger claim. The comparison is single-seed, so with $\pm$ reported across held-out terrains rather than runs the gaps are not statistically resolvable. The 10k budget sits inside the takeoff region, where a shift in takeoff timing moves the endpoint by roughly the observed margin. And the horizon is still a single step, so the target carries little long-range dynamics even though it is now predicted well.

Establishing whether one-step JEPA helps or harms SAC requires matched multi-seed runs at $\geq 100\text{k}$ transitions, with the variance floor and the action-conditioning ablated independently.